Alberta

Wildland / Urban Interface Fires

2015

Government of Alberta

Alberta Emergency Management Agency

A Guide for Municipal Directors of Emergency Management and Consequence Management Officers

Record of Amendments

List of all amendments made to the Guide since inception.

Preface

Fire is a natural part of the ecosystem, and often is necessary for forest renewal and growth. However, since Albertans are increasingly moving into forested regions as a place to live, work, and play, the threat from uncontrolled fires becomes greater. Nowhere is this more apparent than in the Wildland Urban Interface Zone.

The Wildland Urban Interface Zone is that area of Alberta that is neither completely built up, nor completely undisturbed. As such, it is an area that if not properly protected can act as the means of transmission of wildfires from pristine rural areas into heavily populated urban zones, or vice versa.

To combat the threat from wildfires in the Wildland Urban Interface Zone the Government of Alberta has adopted the *FireSmart* Program under the championship of the Ministry of Environment and Sustainable Resource Development. They, along with the Office of the Fire Commissioner and the Alberta Emergency Management Agency, bear the primary responsibility for preventing fires from becoming uncontrollable with consequent destructive impacts.

In the pages that follow, municipal Directors of Emergency Management and Consequence Management Officers will find an overview of how the Government of Alberta manages Wildland Urban Interface fires from the aspects of Prevention and Mitigation, Preparedness, Response, and Recovery. It must be noted that this overview is an introduction to Wildland Urban Interface fires and is intended to supplement, rather than supplant the *FireSmart Guidebook for Community Protection* developed by the Ministry of Environment and Sustainable Resource Development. Greater detail on how to create FireSmart communities, industry sites, and farms is available from the Ministry of Environment and Sustainable Resource Development at http://srd.alberta.ca/Wildfire/FireSmart/FireSmartGuidebookForCommunityProtection.aspx .

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Background

1.1 General

In recent years, the trend of Albertans choosing to live and work in the forested areas of the province has increased. As a result of this, industrial, agricultural, and residential development of previous wildland areas has increased. The area where industrial, agricultural, or residential development overlaps wildland areas is known as the Wildland Urban Interface (WUI), or interface.

The interface presents significant challenges in emergency management and response. Neither fully urban, fully rural, nor unpopulated, it is a zone of overlapping jurisdictions and multiple hazard types. One of the most common hazards in the interface zone in Alberta is fire.

Interface fires can rapidly become devastating. For example, the Flat Top Complex fires of 2011 (also known as the Slave Lake Fire) destroyed 510 buildings in Slave Lake and its surrounding communities, forced the evacuation of 15,000 residents, and to date, has cost over $700 million. In 1919, the Town of Lac La Biche was completely destroyed by an interface fire, killing 14 people.

In order to prevent this type of loss in the future, Alberta will adopt a comprehensive Wildland Urban Interface Fire Strategy designed to ***prevent and mitigate*** the effects of an interface fire**,** ***prepare*** for fire events, ***respond*** effectively before an interface fire can become uncontrollable, and efficiently ***recover*** from the effects of wildfires.

1.2 Definitions

Key definitions are as follows:

a. Ground fires – fires burning in organic soil and decaying woody material beneath the forest floor. They are persistent, slow-burning, and difficult to extinguish.

b. Surface fires – fires burning needles, twigs, branches, bushes, and immature trees on the forest floor. They spread rapidly, but may be contained by firebreaks.

c. Crown fires – fires burning in the upper foliage and branches of mature trees. They are of very high intensity, spread rapidly, and are difficult to contain, even with firebreaks.

d. Firebreaks (also known as fireguards) – barriers to fire spread created by clearing or significantly thinning fuels on a strip of strategically located land. Firebreaks may be linear (a straight line designed to halt the advance of a fire in a particular direction), or ring (a circular firebreak designed to protect infrastructure contained within the ring).

e. Fuelbreaks – trenches dug down to mineral (as opposed to organic) soil. They are designed to prevent the spread of ground and surface fires through denial of fuel. They may be incorporated into firebreaks.

f. Fuels – fuels are all flammable materials, whether natural or man-made.

g. Ladder fuels – bushes, immature trees, and lower branches of mature trees that, once ignited as part of a surface fire, enable the fire to climb mature trees to become crown fires.

h. Wildland – primarily that portion of Alberta defined as the Forest Protection Area (FPA). For the purposes of this strategy, wildland also includes greenbelts in developed areas.

i. Greenbelts – areas of land left in a natural state within a defined developed area.

j. Ministry of the Environment and Sustainable Resource Development (ESRD) – the GOA ministry responsible for the stewardship of air, land, water, and biodiversity in Alberta. It is charged with achieving desired environmental outcomes and sustainable development of natural resources for Albertans.

k. Office of the Fire Commissioner (OFC) – a body within the Public Safety Division of Ministry of Municipal Affairs of the GOA. It is responsible for fire rescue, search and rescue, technical advice on fire response and prevention to Alberta municipalities, certification of firefighting personnel, and public education on fire safety.

l. Alberta Emergency Management Agency (AEMA) – an agency within the Ministry of Municipal Affairs of the GOA. It is the coordinating agency for, and responsible to provide strategic policy direction and leadership to, the GOA and its Emergency Management partners. This role crosses all four pillars of the emergency management framework: Prevention/Mitigation, Preparedness, Response, and Recovery.

1.3 Key References

Forest and Prairie Protection Act – Revised Statutes of Alberta 2000, Chapter F-19.

FireSmart: Protecting Your Community from Wildfire – Partners in Protection, July 2003.

Flat Top Complex: The Final Report from the Flat Top Complex Wildfire Review Committee – ESRD, May 2012.

Firestorm 2003 Provincial Review – Province of British Columbia, February 2004.

An Interface Fire Planning Model: A Case Study of the District of Langford – District of Langford, 2002.

1.4 Framework

The threat of interface fires can be reduced, but never eliminated. In order to minimize the damages to Albertans, the GOA will adopt a WUI Fire Strategy that encompasses the four pillars of emergency management. These pillars are:

a. Prevention / Mitigation.

b. Preparedness.

c. Response.

d. Recovery.

This strategy will enable Albertans to take a cyclical approach to interface fires, rather than focusing on a single incident or fire season, thereby maintaining the strategy's relevance.

Prevention / Mitigation

The best way to combat the effects of a WUI fire are to prevent them from occurring (where possible), and if it is not possible to prevent the fire's ignition, mitigating their severity. This can best be done through a systematic process of risk reduction prior to and during development in the wildland area, as well as through regular upkeep after development is complete. This process is explained below:

2.1 Assessment

The first step in any WUI fire strategy is undertaking an assessment of the risk. Risk is characterized by three elements, only one of which can be influenced by people. The three elements to be assessed are:

a. Fuel - consists of all flammable elements in the interface area, and in the landscape surrounding it. Fuel includes trees, brush, grasses, and any flammable structures built in the zone. This is the one element influenced by people.

b. Weather - Weather effects on fire will influence the behavior of an interface fire. Extended dry periods and high winds will exacerbate fire conditions; conversely, precipitation alone can completely extinguish a fire. While weather cannot be controlled, it is imperative that weather be monitored to properly assess the risk of fire. Historic weather trends in the area must be considered prior to development.

c. Topography - refers to the shape of the land, particularly with respect to slopes. Fire spreads more readily uphill; this needs to be taken into account when siting structures in an interface zone.

Assessment should be conducted prior to any development in an interface zone, and periodically thereafter. Assessment is broken into three major geographic zones as follows:

a. Priority Zone 1 – A 10m bubble around a structure. This is the most critical zone as fire within this zone may come into direct contact with the structure. Fuels in this zone must be carefully managed so that this zone will not support fire of any type.

b. Priority Zone 2 – From 10m to 30m bubble around a structure. This zone may extend further depending on the topography. If the ground is sloping upwards towards the structure, this bubble must be extended down the slope. The primary threat from this zone is flames, radiant heat, and spotting embers. Fuels in this zone must be managed so that any fires in this zone will be of low intensity and slow spread.

c. Priority Zone 3 – From 30m to 100m bubble around a structure. The primary threat from this zone is spotting embers that may be carried to the structure by wind. Fuels in this zone must be managed to minimize the possibility of a high-intensity crown fire.

Assessment in each of these zones includes consideration of additional factors such as the availability of firefighting resources, response times of firefighters, access to structures in the interface zone, and the construction materials used in the structure(s).

Fuel Management is the most time-intensive and most critical element in a successful WUI fire strategy. Once the assessment of the three Priority Zones has been completed, action must be taken to mitigate the risks in these zones. Fuel management consists of the following activities:

a. Removal - Consists of the physical displacement of easily combustible materials away from the structure in Priority Zone 1. This includes surface vegetation such as downed trees, leaves and needles, and shrubs and bushes. Man-made combustible materials such as propane tanks, woodpiles, and outbuildings should also be kept out of Priority Zone 1. Periodic upkeep will be required after the initial removal has been completed.

b. Conversion - Consists of the replacement of highly flammable vegetation with other, less combustible vegetation (environmental factors permitting) in Priority Zone 1. Highly combustible vegetation are those types of plants with a high oil or resin content, those which dry quickly in arid weather, and those which rapidly accumulate large quantities of dead foliage or branches. These types of plants should be replaced if at all possible.

c. Reduction - Consists of lessening the availability of fuels conducive to creating a high-intensity, rapidly spreading fire. It is primarily employed in Priority Zones 2 and 3. Reduction is accomplished through the means of thinning and pruning.

    1) Thinning is the selective removal of trees to reduce the crown cover of the forest, thereby decreasing the rapidity with which a fire can spread.

    2) Pruning is the removal of the understory (defined as vegetation growing under mature trees) and lower branches. The intent of this is to reduce the likelihood that a surface fire is able to climb to the crown.

2.2 Development

The following factors must be taken into consideration as part of the development plan in the interface zone:

a. Access Routes. Ideally, any development in the interface zone should be accessible from multiple directions to prevent becoming isolated by a fire, and be serviced by all-weather roads capable of bearing heavy equipment. These roads should be capable of two-way traffic, incorporate turn-around areas at regular intervals, and be clearly signed. All bridges along the road must be capable of supporting heavy equipment.

b. Aerial Evacuation. For larger developments such as work camps, or isolated developments that are only accessible by one route, aerial evacuation requirements must be taken into consideration. As a minimum, a helicopter landing zone should be identified and prepared, and all necessary equipment (lights, flags) should be maintained at the site.

c. Fire Shelter. In the event of a high-intensity, rapidly spreading fire it may not be possible to evacuate people from isolated areas. Development plans should include at least one fire-resistant structure capable of providing shelter for those people who may be trapped by the fire. This structure should include all necessary supplies (medical, water, food, etc) to permit occupants to survive the fire and shelter in place until such time as a rescue can be mounted.

d. Water Supply. By definition, the interface zone is not serviced by municipal water supplies. Thus, any development in the interface zone must take water supplies that can be used for firefighting into account. These water supplies can be in the form of reservoirs, ponds, storage tanks, or streams, dugouts, and sloughs. Pumps and hoses should be maintained at these locations to permit firefighting in advance of the arrival of professional firefighters. Generators may be necessary to operate the pumps.

e. Firebreaks and Fuelbreaks. Where possible, development in the interface zone should incorporate firebreaks and fuelbreaks.

f. Construction Plan. The manner in which structures are designed and built in the interface zone can greatly reduce the damage likely to be caused by fire. Roofs are of particular importance, and should be constructed of Class ‘A’ rated materials (metal, tile, fiberglass/asphalt composition shingles) as assessed by Underwriters’ Laboratories of Canada where possible. Chimneys and stovepipes must be equipped with a spark arrestor, and have at least 3m spacing from the nearest vegetation. Eaves, vents, and openings should at a minimum be screened with a corrosion-resistant mesh no coarser than 3mm. Siding material (where used) should be as fire resistant as possible (stucco, metal, concrete, etc). Windows and doors should be of tempered glass if possible, and be of double- or triple-pane thickness.

2.3 Weather Forecasting / Fire Monitoring

While it is impossible to control the weather, weather and its potential effects on fires must be closely monitored to properly assess risk. Actively burning fires must also be monitored. Fire weather and fire monitoring can be done at these locations:

http://srd.alberta.ca/Wildfire/FireWeather/Default.aspx

http://srd.alberta.ca/Wildfire/WildfireStatus/Default.aspx

Preparedness

No matter how successful a prevention / mitigation strategy is, over time a fire WILL occur in the interface zone. The degree to which a jurisdiction is prepared for this fire is a key determinant as to whether this will be a fire ***incident***, or a fire ***disaster***. This section details the minimum recommended preparation requirements.

3.1 Mutual Aid Agreements

No jurisdiction can ever have enough resources to address every possible contingency. Mutual Aid Agreements permit authorities to maintain an acceptable level of general firefighting expertise for their jurisdiction, specialize in a particular area of firefighting in terms of training and resources, and through cooperation, provide Albertans with a complete firefighting capability. To be effective, Mutual Aid Agreements should specify, at a minimum, the following:

a. Responsibility and jurisdiction.

b. Equipment inventory and what will be made available as a result of the agreement.

c. Equipment compatibilities.

d. Emergency contact information.

e. Communications protocols.

f. Liability.

g. Cost-recovery procedures and limitations.

Detailed, accurate Mutual Aid Agreements are essential in the Non-Protected Areas (NPA) in Southern and South-Eastern Alberta where the FPA does not apply. Typically ESRD does not pre-position assets in this area; however, extensive 'greenbelts' do present the opportunity for wildland fires for which the more structurally-fire focused municipal fire departments are not necessarily trained or equipped to handle.

3.2 Emergency Planning

The first hours after the start of an interface fire are crucial to a successful response. A well-designed, well-practiced emergency plan can make the difference between success and failure in fighting the fire. Responsible authorities should prepare, maintain, and rehearse a detailed emergency plan that will be immediately effective in fighting the fire while gaining them time to further assess and consider their overall strategy. This emergency plan should as a minimum include the following details:

a. Key contact information including names and phone numbers for all responsible authorities of Mutual Aid Agreements.

b. Standing firefighting resource inventory.

c. Prioritized list of critical infrastructure to be protected.

d. Significant dangerous goods sites.

e. Location and capacities of water supplies.

f. Tactical communications protocols and frequencies.

g. Evacuation protocols to include means of warning, routes, assembly areas, and potential reception centres.

h. Pre-identified firefighter staging areas.

i. Critical infrastructure sprinkler plan.

j. Initial firefighting strategy and objectives.

3.3 Exercises

No plan can be considered as ready to implement until it has been tested. Local authorities, upon completion of their emergency plan, must schedule a series of exercises to test the plan. These exercises should follow a logical progression of ‘crawl-walk-run’ and be so designed as to identify weaknesses and flaws in the emergency plan. Ideally, the series of exercises should culminate in a complete rehearsal of the plan, from alerting to staging firefighting resources to simulated evacuation.

3.4 Cross-training

Structural firefighters approach fires in a completely different manner than wildland firefighting crews, and the converse is also true. Wildland firefighters generally approach a fire through the construction of firebreaks on strategically defensible lines, thereby containing a fire, whereas structural firefighters attack fires more directly to extinguish the fire on structures. As such, they are not readily interchangeable. In order to maximize the employability and effectiveness of firefighting crews, cross-training in both specializations is desirable where possible. The intent of this cross-training is not to completely certify a wildland firefighter as a structural firefighter, nor vice-versa; the intent is to ensure that firefighters are better able to capitalize on each specialty’s strengths, while minimizing their weaknesses.

Response

No matter how thorough the efforts at prevention and mitigation, it is inevitable that interface fires will occur. A timely, decisive response is necessary to ensure that interface fires are reduced and suppressed before they can do substantial damage. Key to the effort of reduction and suppression are well-understood, mutually agreed upon response arrangements.

4.1 Incident Command System

Wildfires, especially those which threaten communities, are characterized by rapidly changing circumstances, confusion (particularly in the fire's initial stages), and the need for a deliberate, well-thought response. In order to impose structure on wildland urban interface fire responses, the GOA uses the Incident Command System (ICS).

Amongst many common characteristics, ICS provides ***Unity of Command, Common Terminology, Management by Objective*, *Flexible and Modular Organization***, and a defined ***Span of Control*** for incident response. The Command and General Staff structure of ICS comprises the following elements:

a. Incident Commander - The Incident Commander (IC) is the individual responsible for all decisions at his or her respective level in regards to management of the incident. The IC is directly assisted by the Public Information Officer, the Safety Officer, and the Liaison Officer who are classified as Command Staff.

b. Operations Chief - The Operations Chief is the individual responsible for executing the tactical operations that fulfill the decisions of the Incident Commander. They control, directly or indirectly (depending upon the scope of the wildfire) all firefighting resources.

c. Plans Chief - The Plans Chief is the individual responsible for collecting and disseminating information on the incident, resources available, and potential resources required. They and their staff (if required) forecast actions and resources that may be needed in the future.

d. Finance and Administration Chief - The Finance and Administration Chief is the individual responsible for tracking all costs and personnel recordkeeping for the incident.

e. Logistics Chief. The Logistics Chief is the individual responsible for providing all resources, services, and support required for the incident.

Greater detail on ICS in wildfire operations is available at:

http://srd.alberta.ca/Wildfire/WildfireOperations/IncidentCommandSystem.aspx

http://www.icscanada.ca/

Training on ICS is available from AEMA at http://apsts.alberta.ca/

4.2 Unified Command

By definition, an interface fire will require response from both wildland firefighters under the control of ESRD and structural firefighters under the control of the Fire Department of local jurisdiction. As a result of their differing approaches to firefighting, Unified Command will be the typical command structure under ICS with respect to the incident. In a Unified Command structure dealing with an interface fire, the *primus inter pares* will be as follows:

a. Fire started in an urban area and is spreading to wildlands – Fire Department of local jurisdiction.

b. Fire started in wildlands and is spreading to an urban area – ESRD.

While it is preferred that the firefighting strategy employed is agreed upon by the two agencies, in the event of disagreement, the primary decision-making authority will be as listed above.

4.3 Resources

Not all firefighting resources are created equal, and great care must be taken to ensure that the proper resource is committed to the proper job and objective. Structural firefighting resources such as sprinklers and ladder trucks must be held for structural fire extinguishing and prevention of fire spread in urban areas, while wildland firefighting resources such as firebreak strike teams and tanker trucks should be held for wildland areas.

4.4 Contractors and Volunteers

Today's fiscal realities make it difficult to maintain a large standing full-time firefighting capability. Thus, maximum use must be made of volunteer firefighting departments and contract firefighting teams, particularly during the peak fire season. The major problems associated with volunteers and contractors are availability and training; municipalities must be proactive in assessing their likely contractor requirements, and must develop and maintain a progressive volunteer training program.

4.5 Federal / Provincial / Territorial / International Assistance

Alberta, through the auspices of the Canadian Interagency Forest Fire Centre (CIFFC), has access to firefighting resources from Canada and the United States. It must be noted that the availability of these resources are dependent upon the fire threat within their home jurisdictions, and should not form an integral element of any emergency plan.

Recovery

Despite the best efforts of firefighters, it may not be possible to prevent loss due to interface fires. The GOA maintains recovery assistance programs designed to return affected communities to pre-disaster conditions.

5.1 Municipal Wildfire Assistance Program (MWAP)

MWAP is a provincial program developed by ESRD and AEMA. It is designed to assist municipalities in both reducing the risk of loss associated with wildfires, and addressing the extraordinary costs of extinguishing wildfires when they occur. Costs eligible under the MWAP include:

a. Costs directly associated with suppression of the wildfire.

b. Regular salary and overtime wages directly associated with suppression of the wildfire.

c. Creation of fireguards.

d. Reclamation of land from fireguards.

e. Restoration of fences cut to create access for firefighting.

Costs NOT eligible under the MWAP include:

a. Those that could be reasonably prevented or avoided.

b. Those for which insurance was readily and reasonably available.

c. Those which likely could be recovered through legal action.

d. Those which could be recovered through other government programs.

e. Those for fires occurring on unoccupied public land.

f. Those for suppression of structural fires.

g. Those for damage to property, such as structures, fences (other than as noted above), or landscaping.

To be eligible for MWAP assistance, the municipality must contact ESRD as soon as practicable and fight the wildfire with ESRD resources if deemed appropriate. The municipality must also investigate the cause of the fire to identify the responsible party, and provide ESRD with a copy of the final investigative report.

In addition to the investigative requirement, the municipality must also have:

a. Signed Mutual Aid Agreements with neighbouring municipalities, industrial fire departments, and ESRD.

b. Included ESRD in its Municipal Emergency Plan.

c. Implemented a fire permit program in accordance with ESRD's Fire Permit Guidelines Manual.

d. Implemented a wildfire awareness program such as FireSmart which encourages residents to take mitigative steps to protect their homes and property from the threat of wildfire.

e. Ensured that one or more senior personnel from the municipal fire department has taken wildfire management courses approved by the local ESRD Wildfire Protection Area Manager.

f. Implemented a proactive aerial and surface infrared scanning program for detection and subsequent action.

5.2 Disaster Recovery Program (DRP)

DRP is a provincial program administered by AEMA by which residents, small businesses, agricultural operations, and municipalities can receive financial assistance to return uninsurable property loss to pre-disaster condition. It must be noted that in order to qualify for a DRP, the event must be considered extraordinary, the effects must be wide-spread, and insurance must not be readily or reasonably available.

Typically, wildfires do not qualify for DRP; however, they may qualify where they pose a threat to urban and commercial developments and then primarily for pre-emptive actions, evacuations, and damaged infrastructure restoration undertaken by government authorities.

5.3 Insurance

The primary means for recovery from wildfire for residents in the interface zone is fire insurance. All people who intend to reside or work in the interface zone should ensure that they have, and maintain, adequate fire insurance to protect their investments.

5.4 Post-Wildfire Activities

A crucial element of recovery activities is investigation to determine the causes of the fire, the effectiveness of the response, and identification of strategies to prevent recurrence or further mitigate the impact of the fire. To this end, after each WUI the following activities will occur:

a. Post-Incident Assessment (PIA). Each organization involved in the WUI fire response will conduct an internal PIA to determine Lessons Learned from the event. In the case ofextraordinary fire events, which would normally have resulted in the elevation of the POC, AEMA will conduct a Cross-Governmental PIA to identify lessons and where applicable, determine best practices involving multiple departments.

b. Action Plan. Based upon the results of the internal PIAs and the Cross-Governmental PIA (if conducted), each organization will develop an action plan to address any shortcomings that were identified.

d. Reassessment. Each organization involved in the WUI fire will conduct a reassessment of the wildfire threat under the new conditions.

Education

There are numerous resources available for people to educate themselves on how to protect themselves from wildfires in the interface zone. One of the most comprehensive resources is the FireSmart Program developed by Partners in Protection. FireSmart offers tips for reduction of the risk of wildland urban interface fires in general, and has separate sub-programs specifically targeting homeowners, communities, agricultural operations, and industry. FireSmart resources are available on the ESRD webpage, located here:
http://srd.alberta.ca/Wildfire/FireSmart/Default.aspx

ESRD also sponsors the FireSmart Community Grant Program which provides municipalities, municipal districts, Métis Settlements, and community associations the opportunity to apply for grants of up to $50,000 for education and risk management projects. Details of the FireSmart Community Grant Program are also available on the ESRD webpage, located here:
http://srd.alberta.ca/Wildfire/FireSmart/FireSmartCommunityGrantProgram.aspx .

Conclusion

Wildfires are, unfortunately, inevitable. However, with proper planning and preparation, the damage caused by wildfires, particularly in the interface zone, can be reduced. In light of the ongoing movement of Albertans to formerly wilderness areas, it is essential that adequate precautions be taken to minimize the impact of wildland urban interface fires. With the adoption of this strategy, the result will be a safer and more secure Alberta.

ANNEX A – Wildfire Preparedness Guide

Community Name

Wildfire Preparedness Guide

INSERT COMMUNITY IMAGE

PREPARED BY

DATE

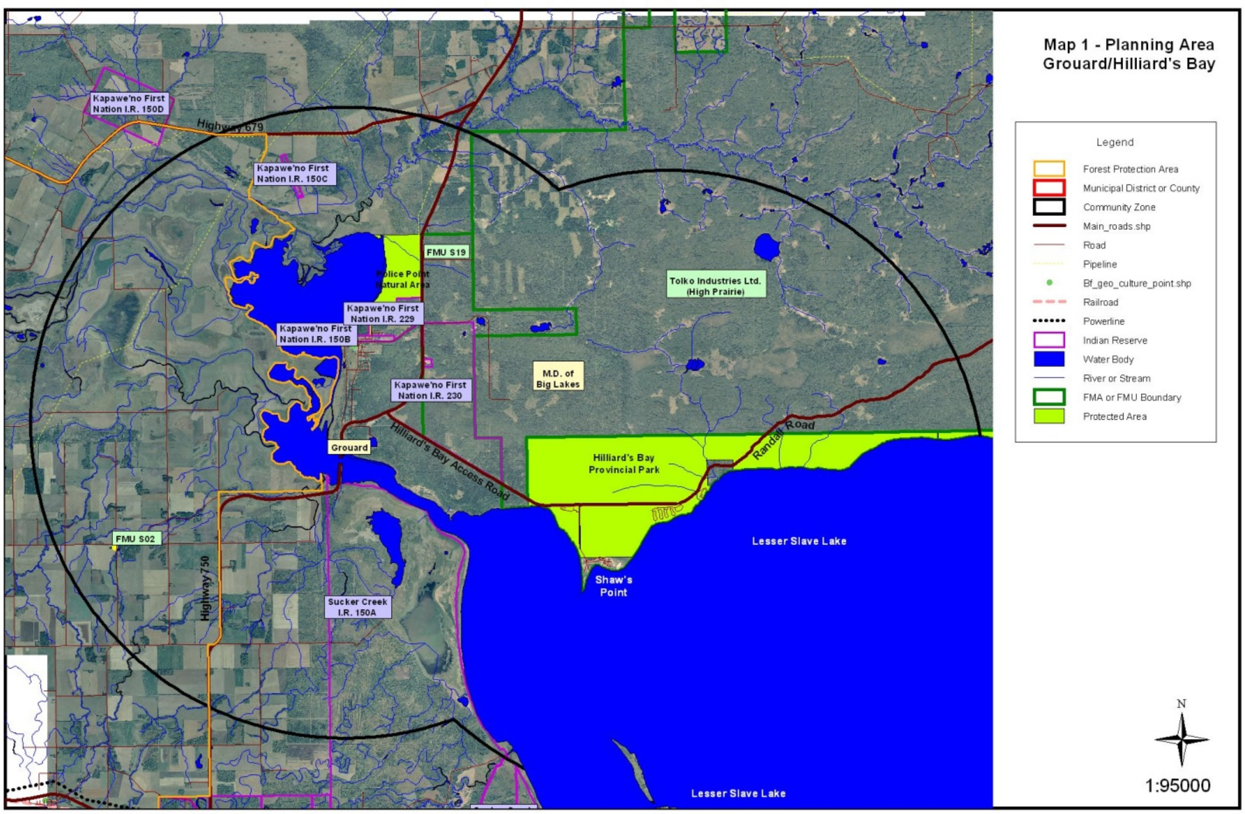
The intent of this document is to provide an Incident Management Team with an overview of the planning area, fire behaviour potential, values-at-risk, and fire operations that can be utilized in the event that a wildfire threatens wildland/urban interface values-at-risk.

WILDFIRE PREPAREDNESS GUIDE

SECTION A: LOCAL AREA DISCRIPTION

A1. PLANNING AREA

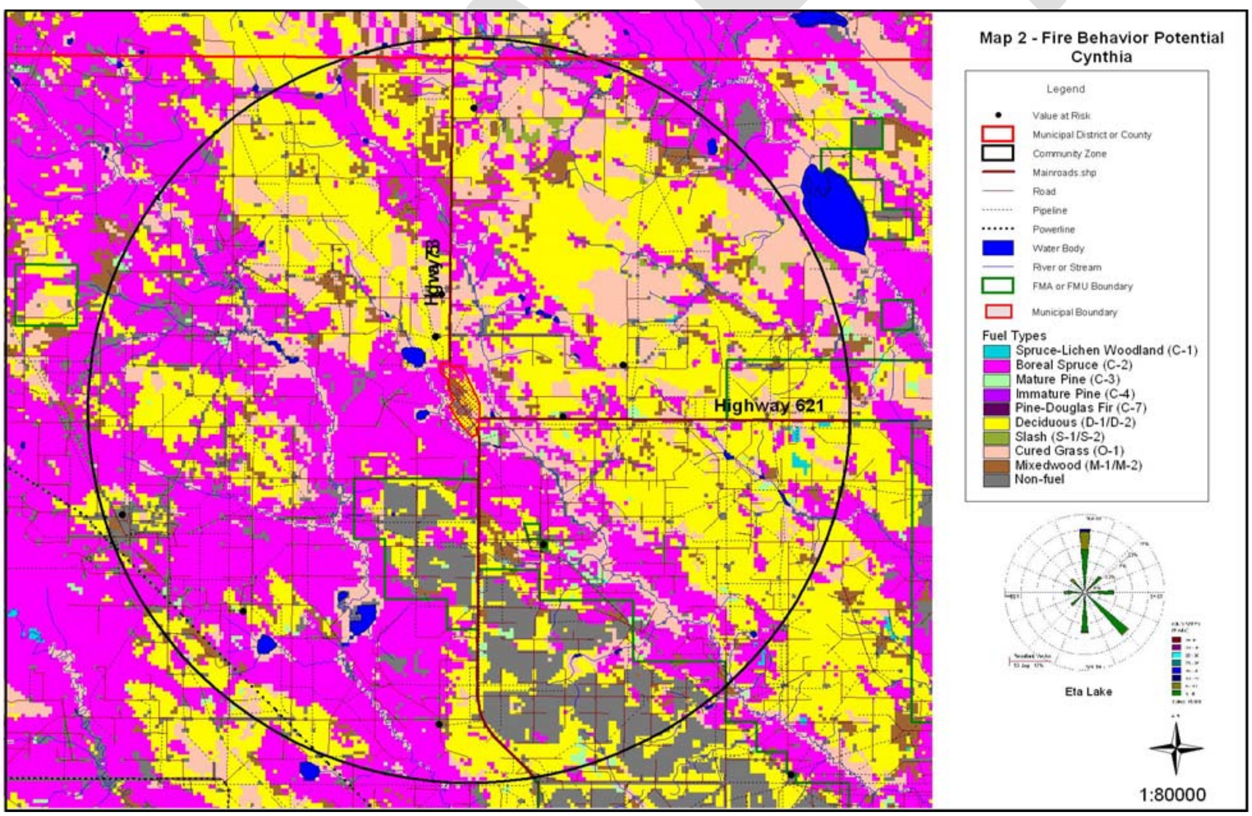
FEATURE	JURISDICTIONAL AUTHORITY
ESRD Management Area	
Municipality(s)	
First Nations	
Forest Management Agreement Area(s)	
Timber Quota Areas	
Provincial Park	



MAP 1: PLANNING AREA MAP (INSERT FOLDED 11 X 17 MAP

A2. FIRE BEHAVIOUR POTENTIAL DESCRIPTION

FACTOR	DESCRIPTION
Fuels Influence	
Weather Influence	
Topographic Influence	
Anticipated Fire Spread	



MAP 2: FIRE BEHAVIOUR POTENTIAL MAP (INSERT FOLDED 11 X 17 MAP)

SECTION B: VALUES-AT-RISK

B1 - STANDARD VALUES-AT-RISK

ID#	Value Name	Value Type	Lat/Long	No. of Structures - Roof Material	
1					
2					
3					
4					
5					

B2 - CRITICAL VALUES-AT-RISK

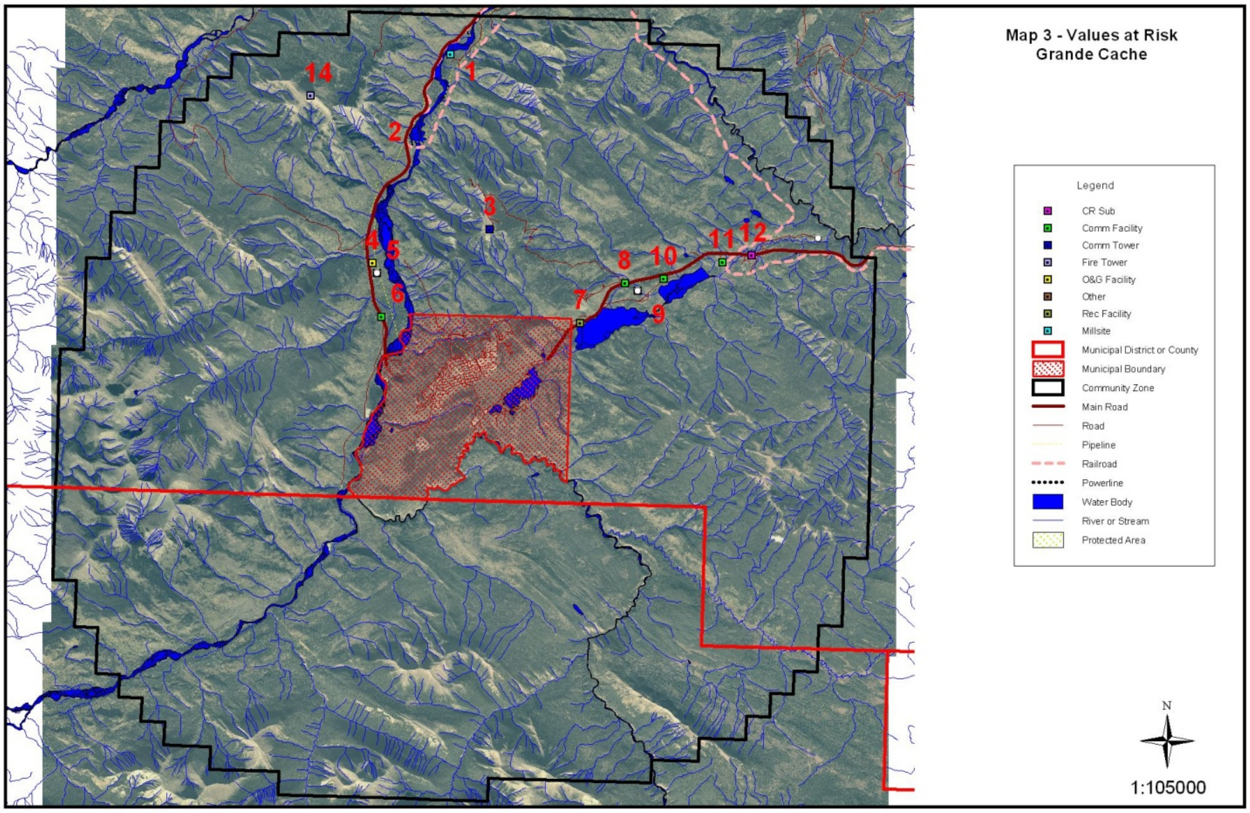
ID#	Value Name	Value Type	Lat/Long	No. of Structures - Roof Material	
1					
2					
3					
4					
5					

B3 - SPECIAL VALUES-AT-RISK

ID#	Value Name	Value Type	Lat/Long	No. of Structures - Roof Material	
1					
2					
3					
4					
5					

B4 - HAZARDOUS VALUES-AT-RISK

ID#	Value Name	Value Type	Lat/Long	Concern
1				
2				
3				
4				
5				



MAP 3: VALUES-AT-RISK MAP (INSERT FOLDED 11 X 17 MAP)

SECTION C: FIRE OPERATIONS

C1 - FUNCTIONAL ROLES	
JURISDICTION	AGENCY RESPONSIBLE
WILDFIRE	
EVACUATION	
VALUES PROTECTION	

C2 - MUTUAL-AID COMMUNICATIONS	
TYPE	FREQUENCY
COMMAND	
TACTICAL	

C3 - MINIMUM AUTO ORDER LIST

ZONE/MAP #	APPARATUS	HEAVY EQUIPMENT	OTHER EQUIPMENT/MANPOWER

C4 - STRUCTURE PROTECTION STRATEGIES & TACTICS

ZONE/MAP #	STRATEGIES	TACTICS	COMMENTS

C5 - FIRE SUPPRESSION WATER SUPPLY

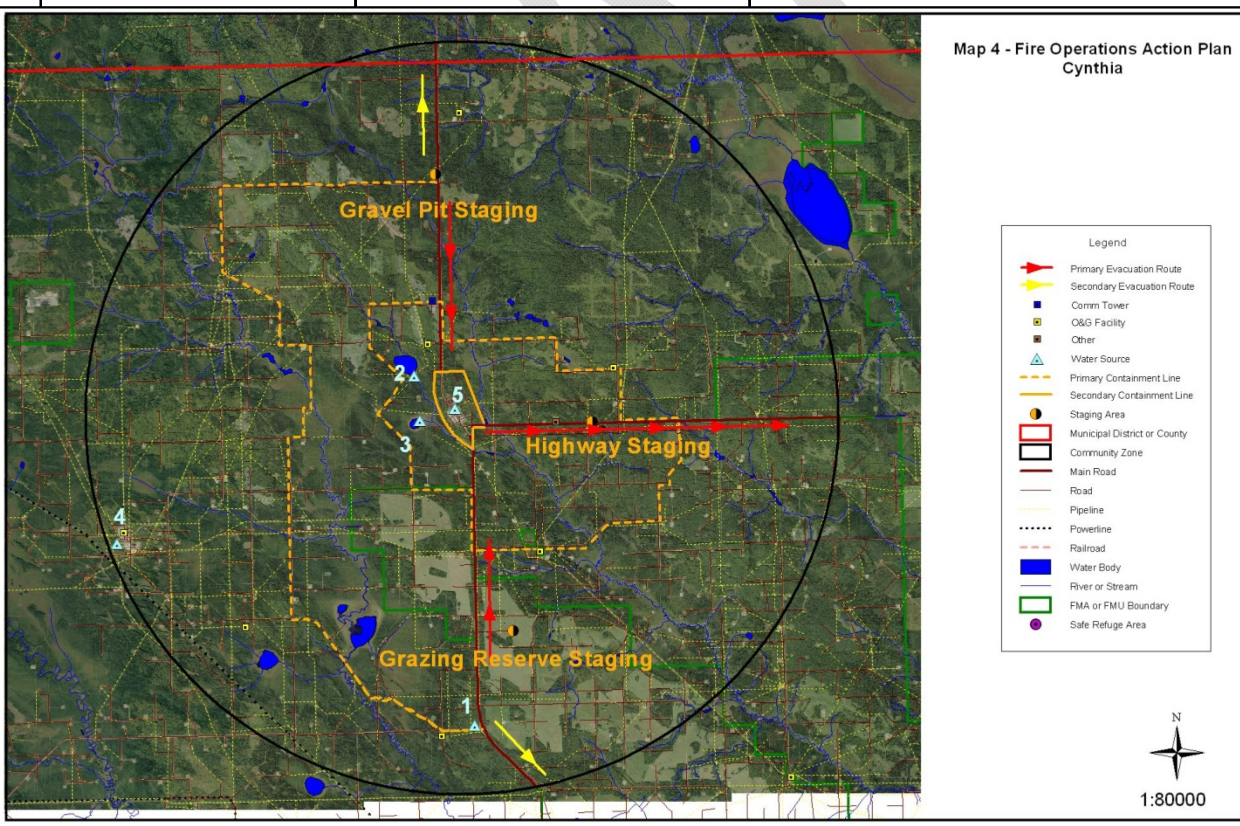
ZONE/MAP #	FACILITY NAME	TYPE	LOCATION	CAPACITY/PRESSURE/FLOW

C6 - STAGING AREAS

MAP #	NAME	LOCATION (LAT-LONG/ATS)	DESCRIPTION
1			
2			
3			
4			
5			

C7 - EVACUATION					
MAP #	VALUE NAME LOCATION	PEOPLE (NO.)	EVAC TIME (MINS)	SAFE REFUGE AREA	SPECIAL EVACUATION CONCERNS
1					
2					
3					
4					
5					

C8 - SAFE REFUGE AREAS			
MAP #	NAME	LOCATION (LAT-LONG/ATS)	DESCRIPTION
1			
2			
3			
4			
5			



MAP 4: FIRE OPERATIONS MAP (INSERT FOLDED 11 X 17 MAP)

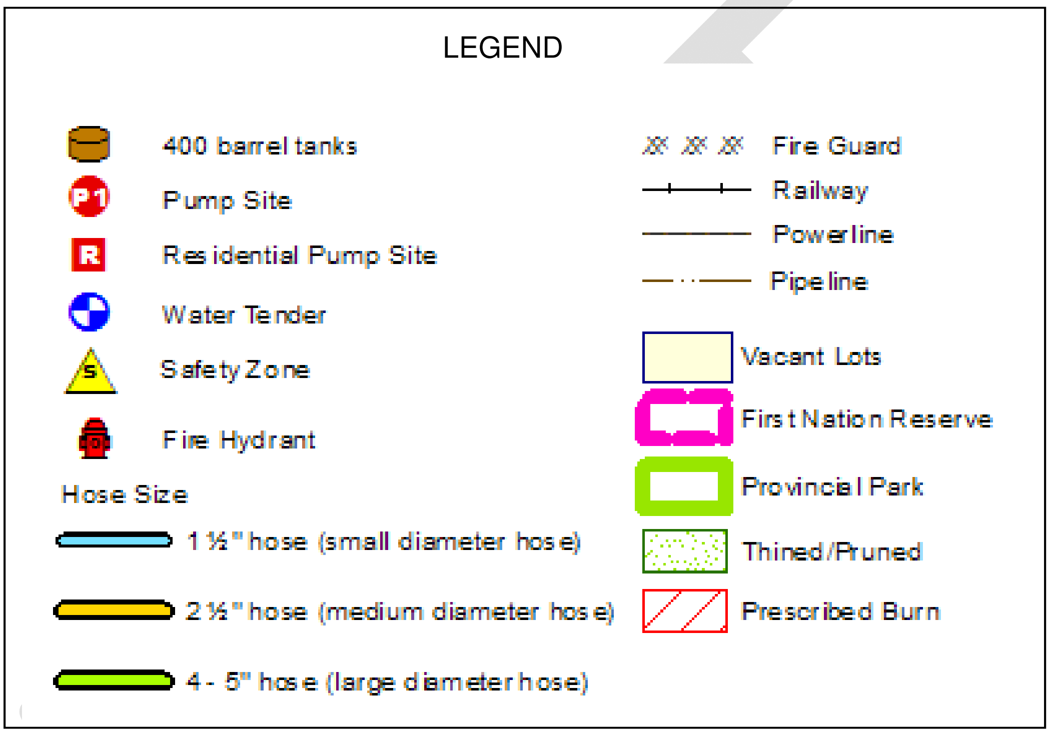
C9 - EMERGENCY CONTACT LIST

EMERGENCY SERVICES				
POSITION	**NAME**	**WORK**	**CELL/PAGER**	**FAX**
FIRE & RESCUE SERVICES				
R.C.M.P				
ESRD - WILDFIRE OPERATIONS				
WILDFIRE MANAGER				
WILDFIRE OPERATIONS OFFICER				
WILDFIRE PREVENTION OFFICER				
PFFC DUTY OFFICER				
AREA DUTY OFFICER				
AREA DISPATCH				
OTHER				

SPRINKLER DEPLOYMENT PLAN

The Calling Lake Sprinkler Deployment Plan will provide the standard.

SCALE: 1:15,000



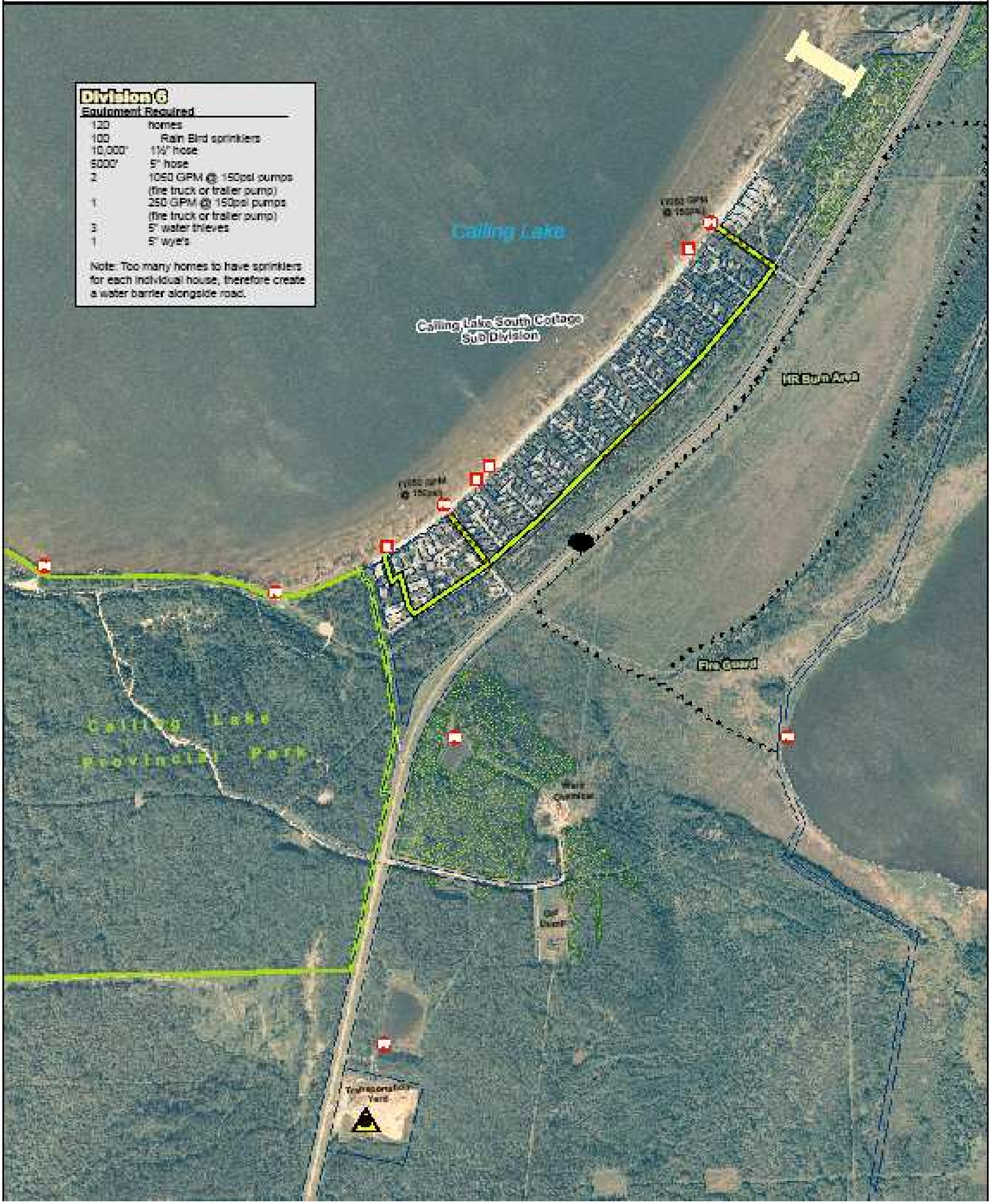
Sprinkler Deployment Plan
Calling Lake

Sprinkler Deployment Plan

Calling Lake

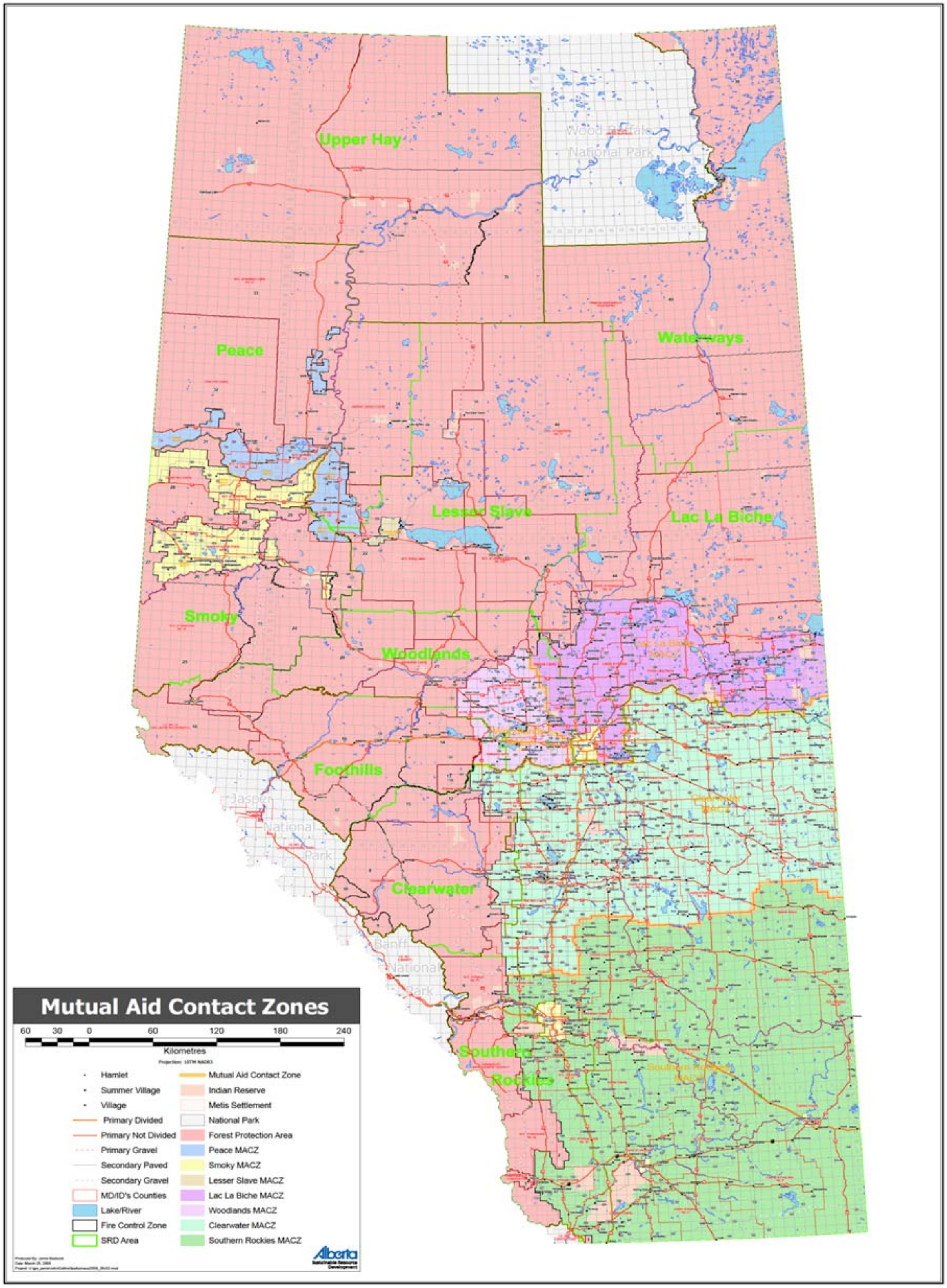
Sprinkler Deployment Plan
Calling Lake

Sprinkler Deployment Plan
Calling Lake



Division 6
Equipment Required
120 homes
100 Rain Bird sprinklers
10,000' 1½" hose
5000' 5" hose
2 1050 GPM @ 150psi pumps (fire truck or trailer pump)
1 250 GPM @ 150psi pumps (fire truck or trailer pump)
3 5" water thieves
1 5" wye's
Note: Too many homes to have sprinklers for each individual house, therefore create a water barrier alongside road.
Calling Lake
1050 GPM @ 150psi
Calling Lake South Cottage Sub Division
1050 GPM @ 150psi
HR Burn Area
Fire Guard
Calling Lake Provincial Park

ANNEX B – Mutual Aid Contact Zones



ANNEX C – Consequence Management Officer Responsibilities At the POC

CMO Responsibilities At the POC					
	Attendance at POC (Most likely - requirements may change with circumstances)				
Department	Level 2	Level 3	Level 4	Responsibilities	Remarks
Aboriginal Relations		Yes		- Liaise with Aboriginal Affairs and Northern Development Canada - Liaise with non-treaty aboriginal settlements	- Only required if wildfire impacts aboriginal and Metis settlements
Innovation & Advanced Education		Situationally Dependent		- Provide advice on Enterprise & Advanced Education infrastructure that may be affected by the wildfire	- Only required if wildfire impacts Enterprise & Advanced Education infrastructure
Agriculture & Rural Development		Yes	All	- Provide advice on carcass disposal - Provide advice on general agricultural impacts of wildfire	
Culture and Tourism		Situationally Dependent		- Provide advice on cultural infrastructure that may be affected by the wildfire	- Only required if wildfire threatens cultural sites
Education		Situationally Dependent		- Provide advice on on schools that may be used for emergency activities	
Energy		Situationally Dependent		- Provide advice on Energy infrastructure that may be affected by the wildfire	

AER		Yes		- Liaise with industry leaseholders affected by wildfire - Provide advice as to wildfire impact on industry infrastructure	
ESRD	Yes	Yes		- Provide interface between tactical firefighting operations and POC - Provide regular updates on fire behaviour and prognosis in the Forest Protection Area - Provide interface between POC and people at affected provincial parks - Make provincial parks available for emergency activities as required	
Treasury Board & Finance					- Treasury Board generally is required after the crisis is resolved, and thus is not needed at the POC until later
Health		Yes		- Provide advice on health effects of reduced air quality caused by wildfires - Provide advice on wildfire impacts to patients at affected hospitals	
Human Services		Yes		- Provide regular updates on reception centre registrations and activities	
Infrastructure		Situationally Dependent		- Provide advice on Governmental facilities that may be used for emergency activities - Provide advice on Business Continuity impacts on Government infrastructure affect by wildfire	- May be required for Level3, dependent upon wildfire location

Intergovernmental & International Relations		Situationally Dependent		- Liaise with adjoining jurisdictions	- Only required if GoA response involves other jurisdictions
Justice & Solicitor General		Yes		- Provide interface between tactical operations and the POC - Provide advice on potential road closures - Provide advice on potential evacuation routes	
Municipal Affairs	Yes	Yes		- Provide advice to affected municipalities	- Most Municipal Affairs functions will be executed by AEMA Field Officers
Office of the Fire Commissioner	Yes	Yes		- Provide interface between tactical firefighting operations and POC - Provide regular updates on fire behaviour and prognosis in affected municipalities	
Public Affairs Bureau		Yes		- Provide interface between the POC and the media - Provide Subject Matter Expertise on all messaging from the POC	
Service Alberta		Yes		- Provide interface between POC and vendors for emergency contracting	
Transportation		Yes		- Coordinate availability of provincial transportation means for movement of emergency vehicles - Liaise with operators of non-provincially controlled transportation infrastructure for movement of emergency vehicles - Coordinate emergency repair for transportation infrastructure as needed	

Notes:

1. This list of responsibilities does not include tactical actions taken at the scene. This list only refers to the CMO at the POC.

2. The primary duty of all CMOs is to act as the liaison between their departments and the POC.

3. The Lead Agency for Wildfire Urban Interface fires will be ESRD, OFC, or Unified Command between the two.

4. The Lead Agency for Coordination will be AEMA.

5. This list of responsibilities is not exhaustive. Additional requirements will be situationally dependent.

6. Attendance at the POC will be situationally dependent. Some CMOs may be tasked to work remotely.